Project Planning

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

This plan breaks the project into phases with deliverables, effort, and risks. Durations are expressed in weeks from project start so the plan stays valid whatever the calendar dates; adjust them to your institution's schedule.

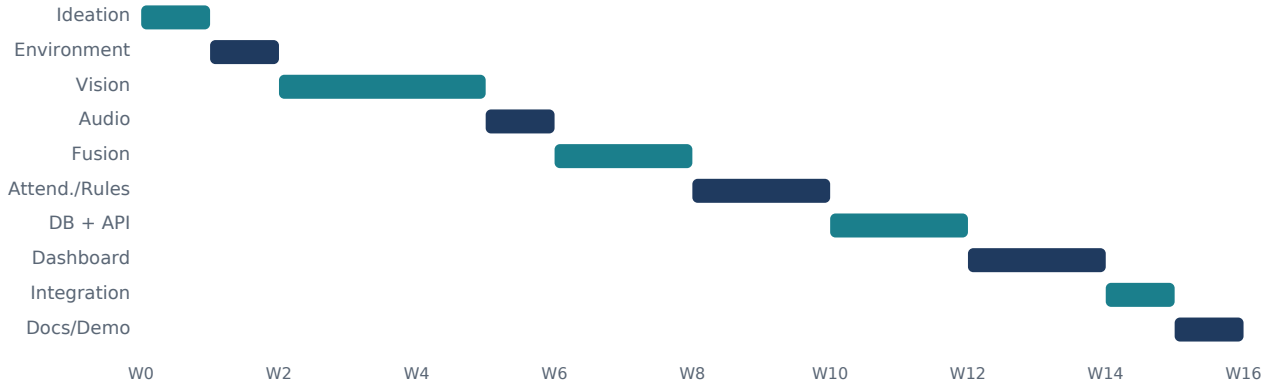
1. Project Overview

Item	Detail
Project	Multimodal Attendance & Engagement Agent
Team / owner	Nirbhay Dubey, B.Tech Computer Science & Engineering (Year 1)
Goal	Offline AI system for automatic attendance and engagement tracking with teaching recommendations
Method	Incremental build: one service at a time, each verified by its own test suite
Final status	All 12 automated test suites passed; audit and handover document completed on October 19, 2026

2. Work Breakdown and Timeline

Phase	Weeks	Key tasks	Deliverable / exit check
1. Ideation and requirements	1	Brainstorm, define problem, empathy map, requirements	Approved scope and requirement list
2. Environment and test media	2	Python 3.11 venv, install libraries, generate synthetic video/audio	verify_env.py passes; test_classroom.mp4 / .wav ready
3. Vision pipeline	3-5	Camera service, Haar detector, DeepSort tracker, recognition, head pose and gaze	test_face_detection, test_face_recognition, test_head_pose_gaze pass
4. Audio pipeline	6	Feature extraction, noise floor, VAD classifier	test_audio_analysis, test_vad_noise_classifier pass
5. Fusion and scoring	7-8	Temporal sync buffer, engagement engine, EMA smoothing	test_multimodal_alignment, test_engagement_engine pass
6. Attendance and reasoning	9-10	Attendance service, manual override, 5-rule recommendation engine	test_attendance_service, test_reasoning_engine pass
7. Database and API	11-12	SQLAlchemy models, CRUD, FastAPI REST + MJPEG + WebSocket	test_database, test_api_endpoints pass
8. Dashboard	13-14	Streamlit 8-page UI, camera toggle, exports, settings	Manual walkthrough of all pages
9. Integration and testing	15	End-to-end 11-step pipeline test, bug fixes	test_end_to_end_pipeline passes
10. Documentation and demo	16	Technical audit, viva questions, demo script	Handover document, demo rehearsal

Timeline (weeks)



Relative schedule; shift it to match the actual semester calendar.

3. Milestones

Milestone	Success criterion
M1 Perception ready	Faces detected, tracked, recognized; pose and gaze estimated on test video
M2 Multimodal fusion ready	Audio and vision aligned; engagement tiers computed
M3 Backend and database ready	Sessions and attendance saved; WebSocket streams telemetry
M4 Dashboard complete	All 8 pages working against the backend
M5 Release candidate	12 of 12 test suites pass; demo script rehearsed

4. Roles and Responsibilities

Role	Responsibility
Project owner / developer	Design, implementation, testing, documentation, and demo
Faculty evaluator	Review scope, assess milestones, conduct the viva
AI engineering assistant	Code review, test generation support, and technical audit

5. Risk Register

Risk	Likelihood	Impact	Mitigation
No camera available during demo	Medium	High	Synthetic test_classroom.mp4 fallback built into CameraService
Poor lighting reduces recognition	Medium	Medium	Conservative threshold; anonymous track fallback; lecturer override
SQLite lock contention at 30 fps	High	Medium	Keep per-frame metrics in memory; commit summaries only at session end
No microphone listener in backend	Medium	Low	Audio defaults to QUIET; chunks can be posted to /api/audio/chunk
Privacy objections	Medium	High	Offline design, consent enforcement, no raw video stored
Single camera limits coverage	High	Medium	Document limitation; plan multi-camera RTSP as future work

6. Tools and Resources

- Development: Python 3.11, VS Code or similar, Git.
- Libraries: OpenCV, MediaPipe, DeepFace, DeepSort, Librosa, FastAPI, SQLAlchemy, Streamlit.
- Hardware: standard laptop with webcam; no GPU needed.
- Test data: generated with generate_test_media.py (synthetic video and audio).